

Food factories

Plants are unique – they are the only living things that make their own food. This process is called photosynthesis, which can only happen in the daytime because it uses sunlight.

What's cooking?

Leaves take in sunlight, carbon dioxide (CO_2) from the air, and water from the roots and mix them together using chlorophyll. This recipe makes glucose, a type of sugary plant food, and oxygen.

Air holes

The carbon dioxide used in photosynthesis is taken in through stomata, tiny holes on the surface of every leaf. They also “breathe out” any extra oxygen that the plant does not use for respiration.

Leaf words

Photosynthesis The unique way plants make their own food.

Respiration How living things turn food into energy.

Chlorophyll A chemical found inside leaves that is used for photosynthesis.

Going green

This close-up of a leaf shows green dots inside the cells.

These are the stores of chlorophyll, and they give leaves their green colour.

Carbon dioxide

Oxygen

Plants lose water and gases from their leaves.

Sunlight

Open stomata